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# ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)

**ORGANISATION OF ISLAMIC COOPERATION (OIC)**

**Department of Computer Science and Engineering (CSE)**

**QUIZ 4 SET B** **WINTER SEMESTER, 2024–2025**

**DURATION: 30 MINUTES** **FULL MARKS: 20**

# Math 4341: Linear Algebra

Programmable calculators are not allowed. Do not write anything on the question paper.
Answer **all 2 (two)** questions. Figures in the right margin indicate full marks of questions with corresponding COs and POs in parentheses.

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1. Find the Singular Value Decomposition (SVD) ($A = U\Sigma V^T$) for $A = \begin{bmatrix} 1 & 0 & 1 \\ -1 & 1 & 0 \end{bmatrix}$ 10 (CO3) (PO1)

2. A sequence is given by 10 (CO3) (PO1)

$$1,\ 4,\ 5,\ -2,\ -19, \cdots$$

where each term is obtained as a linear combination of the two preceding terms. The first two terms of the sequence are $a_0 = 1$ and $a_1 = 4$.

   a) Determine the recurrence relation for the sequence. In other words, express $a_n$ in terms of $a_{n-1}$ and $a_{n-2}$.

   b) Express the recurrence relation obtained in Question 2.a in matrix form and use diagonalization to derive a formula for the $k$-th term of the sequence.

   c) Using the result obtained in Question 2.b, compute the $10^{\text{th}}$ term of the sequence and verify that $a_{10} = -307$.